

HUAI PONG AGROMET, Thailand

WMO: 484790

Lat: 12.7351N

Lon: 101.1353E

Elev: 45

StdP: 100.78

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 20.0	(c) 21.2	(d) 11.4	(e) 8.4	(f) 22.9	(g) 13.4	(h) 9.6	(i) 24.7	(j) 5.1	(k) 26.1	(l) 4.5	(m) 26.3	(n) 1.6	(o) N/A	(p) 0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 5.9	(c) 34.5	(d) 26.8	(e) 33.9	(f) 26.7	(g) 33.3	(h) 26.4	(i) 27.9	(j) 32.5	(k) 27.6	(l) 32.1	(m) 27.4	(n) 31.7	(o) 1.3	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.7	22.4	30.2	26.5	22.1	30.1	26.2	21.7	29.9	89.4	32.6	88.4	32.4	87.2	32.0	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
4.2	3.4	2.9		17.8	36.0	1.7	0.6	16.6	36.4	15.6	36.8	14.7	37.1	13.4	37.6	
				13.7	28.7	2.0	0.4	12.2	29.0	11.0	29.3	9.9	29.5	8.4	29.8	
				DB												
				WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.5	26.8	27.8	28.9	29.9	29.9	29.4	29.0	28.9	28.4	27.9	27.9	27.0
	DBStd	1.62	1.51	1.37	1.39	1.58	1.40	1.11	1.09	1.02	1.07	0.98	1.12	1.67
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6745	522	499	585	598	616	581	591	584	552	554	536	528
	CDD18.3	3703	264	266	327	348	357	331	332	326	302	296	286	270
	CDH23.3	41631	2460	2736	3802	4420	4585	4102	3994	3833	3334	2936	2878	2552
CDH26.7	16603	842	957	1532	2153	2172	1777	1630	1501	1168	931	1022	919	
Wind	WSAvg	0.6	0.7	0.5	0.6	0.5	0.4	0.5	0.6	0.5	0.6	0.6	0.8	1.1
Precipitation	PrecAvg	1352	26	26	63	76	170	168	153	129	257	220	51	9
	PrecMax	1798	100	83	245	195	342	331	298	306	443	382	126	41
	PrecMin	949	0	0	1	2	53	19	32	42	125	82	0	0
	PrecStd	206	29	22	53	48	77	76	84	59	75	71	38	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.4	34.1	35.7	35.2	34.7	34.0	33.9	33.6	33.1	33.3	33.6
		MCWB	23.2	24.2	26.4	26.4	27.2	27.6	26.5	26.5	26.3	26.2	24.6	24.2
	2%	DB	32.1	32.5	33.1	34.6	34.3	33.7	33.2	32.9	32.7	32.2	32.7	32.6
		MCWB	23.3	24.6	26.2	26.9	27.2	27.2	26.5	26.6	26.4	26.1	24.8	24.3
	5%	DB	31.4	31.7	32.4	33.8	33.6	32.9	32.6	32.2	32.0	31.6	32.0	31.8
		MCWB	23.5	24.8	26.0	26.9	27.2	26.9	26.4	26.5	26.3	26.1	25.1	24.1
	10%	DB	30.6	30.9	31.7	33.0	32.8	32.2	31.8	31.6	31.2	30.8	31.3	30.9
		MCWB	23.6	24.7	25.9	26.7	27.0	26.8	26.4	26.3	26.2	26.0	25.0	23.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	26.7	27.2	28.1	28.2	28.2	27.7	27.7	27.7	27.6	27.2	27.0
		MCDB	30.4	31.3	32.4	33.2	32.9	33.0	32.2	31.8	31.8	31.2	31.2	31.2
	2%	WB	25.8	26.3	26.8	27.7	27.9	27.7	27.2	27.3	27.2	27.1	26.7	26.1
		MCDB	29.5	30.5	31.5	32.5	32.4	32.2	31.5	31.3	31.0	30.8	30.5	30.4
	5%	WB	25.4	26.0	26.6	27.4	27.6	27.4	26.9	27.0	26.9	26.7	26.2	25.5
		MCDB	29.0	30.0	31.0	32.1	32.0	31.6	31.1	30.9	30.5	30.2	30.0	29.5
	10%	WB	25.0	25.6	26.2	27.1	27.3	27.1	26.6	26.6	26.5	26.3	25.8	25.0
		MCDB	28.4	29.3	30.3	31.7	31.4	31.0	30.5	30.3	29.9	29.5	29.4	28.8
Mean Daily Temperature Range	5% DB	MDBR	8.3	7.0	6.1	6.3	5.9	5.7	5.6	5.8	5.7	6.2	7.1	8.1
		MCDBR	9.2	8.0	7.0	6.9	6.6	6.7	6.6	6.6	6.8	7.5	8.1	8.9
		MCWBR	3.8	2.9	2.2	2.1	2.1	2.4	2.2	2.3	2.4	2.9	3.1	3.5
	5% WB	MCDBR	7.1	6.4	6.2	6.5	6.1	6.2	6.0	6.2	6.3	6.6	7.1	7.7
		MCWBR	3.2	2.4	2.1	2.4	2.3	2.5	2.3	2.6	2.6	3.0	3.1	3.2
Clear-Sky Solar Irradiance	taub	0.535	0.588	0.595	0.538	0.475	0.458	0.456	0.456	0.466	0.530	0.491	0.478	
	taud	2.007	1.885	1.913	2.090	2.317	2.383	2.383	2.370	2.326	2.109	2.203	2.202	
	Ebn at Noon	768	746	752	791	827	832	837	847	845	784	804	810	
	Edh at Noon	173	203	201	168	131	121	122	125	131	160	141	140	
All-Sky Solar Radiation	RadAvg	5.03	5.43	5.55	5.82	5.19	4.98	4.82	4.94	4.78	4.77	4.95	4.93	
	RadStd	0.33	0.28	0.41	0.38	0.34	0.27	0.29	0.33	0.28	0.33	0.37	0.26	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.16	N/A	N/A	+72	+72

Nomenclature: See separate page